ANALYTICS GROUP PROJECT

OBJECTIVE

Use real world data (e.g., Minnesota Pollution Control Agency (MPCA) provided data sets) to answer the given questions & share insights of your findings.

DATA OVERVIEW

The data was provided by the Minnesota Pollution Control Agency (MPCA). There are three datasets related to recycling in all MN counties, including materials, residential recycling, commercial recycling, weight of different wastes, expenditure and revenue related to recycling. The names of the datasets are "Recycling by category", "Waste management by method", and "Expenditure and revenue". The history of data collection, relevant recycling policy changes, and a detailed description of variables are provided in the following sections.

Recycling Data Collection History in MN

In 1989, the Governor's Select Committee on Recycling and the Environment (SCORE) recommended to the Legislature to adopt a comprehensive set of laws, commonly referred to as SCORE. This act initiated a state funding source for recycling programs, as well as waste reduction, management of household hazardous wastes, and problem materials. SCORE legislation and grant dollars, along with funding from counties and local government, provide the basis for long-term, flexible programs.

The first SCORE report which was published in December of 1990 represents October 4, 1989 through June 30th 1990. SCORE is a long-term funding mechanism for the counties that are based on population (except a county may not receive less than $55,000 in a fiscal year). There is also a list of permissible expenditures as outlined in 115a.557 subd. 2 and counties must match at least 25% of SCORE funds with local revenue.

The SCORE annual report summarizes information submitted by all 87 counties and the Western Lake Superior Sanitary District on waste management efforts, including funding and costs, waste reduction activities, recycling, composting, household hazardous waste programs, and problem materials collection. This information is used to calculate the cost of managing waste and recycling.

Changes in SCORE

1993: Source reduction credit added to 1996 goals:

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Sec. 13. Minnesota Statutes 1992, section 115A.551, subdivision 2a, is
amended to read:
     Subd. 2a. [SUPPLEMENTARY RECYCLING GOALS.] By December 31, 1996, each
     county will have as a goal to recycle the following amounts:
     (1) for a county outside of the metropolitan area, 30 percent by
         weight of total solid waste generation;
     (2) for a metropolitan county, 45 percent by weight of total solid
         waste generation.
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1995: Yard waste credit added to 1996 goals:

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Subd. 2a. [SUPPLEMENTARY RECYCLING GOALS.] (a) By December 31, 1996, each
county will have as a goal to recycle the following amounts:
     (1) for a county outside of the metropolitan area, 30 35 percent by
         weight of total solid waste generation;
     (2) for a metropolitan county, 45 50 percent by weight of total solid
         waste generation.
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2014: The legislature updated state government's recycling goals, requiring all state agencies in the Twin Cities metro area to achieve a recycling rate of 75 percent or higher by the year 2030.

2015: Addition to the definition of recyclable materials
In 2015, legislation was passed to encourage organics recovery. Organics were separated from other recycling.

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Subd. 25a. "Recyclable materials" means materials that are separated from
mixed municipal solid waste for the purpose of recycling or composting,
including paper, glass, plastics, metals, automobile oil, batteries, source-
separated compostable materials, and sole source food waste streams that are
managed through biodegradative processes.
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Descriptions of Variables

In "Recycling by category" data:

- Year: year of data collected
- County: county name in MN state
- Category: category of recycling
- Material: materials in each recycling category
- Res Tons: Residential recycling weight in tons
- CII Tons: Commercial, institutional and industrial recycling weight in tons
- Total Tons: Total recycled weight in tons at each county

In "Waste management by method" data:

- Year: year of data collected
- County: county name in MN state
- Method Recycling: recycled materials in tons at each county
- Method Organics: organic wastes in tons at each county
- Method Onsite: wastes disposed on physical properties: waste burned or buried at properties
- Method WTE: waste to energy: wastes burned to energy
- Method Landfilled: wastes buried underground of the 21 municipal composting sites

In "Expenditure and revenue" data:

- Year: year of data collected
- County: county name in MN state
- Expenditure Administration: administration cost of recycling
- Expenditure Education: educational cost of recycling
- Expenditure HHW Problem Material Mgmt: costs on management of household hazardous waste (HHW) which are problem materials in the municipal solid waste stream. Examples of those materials are oil-based paint, weed killers and etc.
- Expenditure Organics: costs on management of organics
- Expenditure Recycling: costs on management of recycling materials
- Expenditure Source Reduction: costs on upstream activities (e.g. educations) targeted at reducing the volume, weight or toxicity of waste generated
- Expenditure Waste-to-Energy: costs on management or education activities related to waste-to-energy
- Revenue SCORE: funds obtained from state or federal for SCORE
- Revenue Local Revenue: grants obtained from landfill and processing
- Revenue Other Revenue: adjustments carried over from previous years; HHW funding from MPCA or others; material sales and others.

Note that all three data sets are available at the D2L
(Materials → Content → Assignment → Group Assignment)

QUESTIONS TO ANSWER

Question 1: Are there any different recycling patterns between counties for each type of waste (organics, WTE, landfilled etc.) based on the *waste management by method* data? Are the patterns consistent for different types of wastes?

Question 2: Are there any different recycling patterns between counties for each type of recycling category (paper, glass, metal etc.) based on the *recycling by category* data? Are the patterns consistent for different types of recycling category?

Question 3: What is the average expenditure by category by year, visually and statistically? Is the expenditure by category statistical meaningful? If so, what did you discover?

Question 4: Does the net income (revenue - expenditure) have any impacts on recycling? Is there any correlation between net income and the amount of total recycling?

Question 5: Perform appropriate visualization to examine patterns in the data related to policy changes in the history of recycling. Part of the history of policy changes were explained in the data overview section. Do research on other possible policy changes matching your findings.

Question 6: Do your own research on national recycling data and find the national pattern of recycling (please do not forget about explaining your dataset and source of it). Is that consistent with Minnesota's pattern?

GRADING SCHEME

Your submission will be evaluated based on the following rubrics;

Category	**Superior**	**Very Good**	**Need improvement**	**Emerging**
Completeness and breadth	The analysis addressed all questions and the analysis was thorough and appropriate	The analysis addressed most of the questions and the analysis was thorough and mostly appropriate	The analysis addressed about half of the questions and the analysis was not very appropriate	The analysis addressed less than half of the questions and the analysis has major flaws
Appropriateness of analytical methods	The analysis methods were sophisticated, well explained and correct	The analysis methods were well explained, but not detailed. The methods were correct	There were minor flaws of the analysis methods, and not well explained	The analysis had major flaws and not well explained
Creativity and innovation	The analysis and visualization were extremely creative. The insights and recommendations of results provided were very innovative	The analysis and visualization were somewhat creative. The insights of results provided were innovative	The analysis and visualization were at baseline and conventional with some innovative insights of results provided	The analysis and visualization were not creative and the insights of results were conventional
Communication of outcomes	Methods and results were conveyed in a detailed manner. Clearly explained the outcomes and provided clear and relevant insights of the outcomes	Methods and results were conveyed in a manner that is easily understood. Clearly explained the outcomes. Insights were provided but not very relevant	The description of methods was not complete. The explanations of outcomes were clear. Not much insights provided related to the outcomes.	The description of methods was minimal. The explanations of outcomes were not very clear. No insights provided
Data Processing	Data resources were appropriate and relevant. Data processing related to missing values, merging, row and column manipulation and etc. was very appropriate and well addressed	Data resources were appropriate and relevant. Data processing related to missing values, merging, row and column manipulation and etc. was appropriate but not well addressed	Data resources were appropriate and relevant. Data processing related to missing values, merging, row and column manipulation, etc. was not appropriate	Data resources were not appropriate and relevant. Data processing related to missing values, merging, row and column manipulation and etc. was not addressed
Team synergy	The team demonstrated a high level of coordination, cooperation and synergy, and all team members participated in the presentation	The team demonstrated some level of coordination, cooperation and synergy, and most team members participated in the presentation	The team lacked coordination of team work and very few team members participated in the presentation	No coordination of team work, and only one team member participated in the presentation